

Senior Project Defense

CSCI 499

FitnessJust4You! Fitness Web Application

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Bachelor of Arts in Computer and Information Science

By

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Statement of Purpose:

The main purpose of creating the type of project that I did was to create something that could possibly be beneficial to others, if I planned to use it for more than just my senior project. This project is important because there are a lot of people who would like to keep track their calorie intake, but either don't know how to or they may not have the resources to log it down. Obesity has been an ongoing issue that has been addressed and there are individuals taking the steps that they need to handle the situation, but still might need some help, I believe this website could help individuals learn how many calories they are consuming with what they are eating on a regular basis. The website can also recommend them with some better food choices. But obesity is not the only issue that we are worried about, this website also has a section for individuals who are looking to gain weight.

Of course, this will be done by continuing to eat healthy. The way that we would encourage our users to gain weight would not be by forming unhealthy diets and adding on fat. Instead by a sticking to a specific diet and working out, which will help that user pack on body mass vs body fat. The website will also provide workouts that the user would want or need in order to either lose weight, sustain their weight, or even gain weight. There will three different options of workouts that the user can chose (light, moderate, or intense) from based on either what they feel like doing that day or what they think they can physically handle. I believe if I were to continue improving this app and allow it to go public that it could help and affect a lot of people lives.

Research/Background:

During this project I have had to do a lot of research in order to complete the tasks at hand. I have had some background in this area, that I have learned though the Applied

Networking course that I have taken here at Charleston Southern University, but the knowledge that I needed to complete this project was more extensive than the knowledge I have previously gained from the class. My knowledge was of the basics as far as coding in HTML and incorporating CSS into that code to make it look more user friendly and C# to code the server side. That however was not enough, I have had to go above and beyond to research, learn, and teach myself how to incorporate bootstrap in to program in order to make the visuals of the website like a professional, fully functional website. I have also had to re-learn how to properly set up an SQL database and connect it to the asp.net page. All of these features have required a lot of research, but it was great doing so, as I have learned a lot through my studies and research.

Hardware/Software Required for Project:

The hardware that was required for this project was the server (my laptop), power supply units, and an external hard drive (purchased after the third computer failure). The software that was needed for the project, has a more extensive list. First and foremost, I used the visual studios 2017 software in order to create my website. The bootstrap4 and CSS software was used to make the website look more presentable. I also used JAVASCRIPT and C# in order to write the server-side code. This software brought the visuals of my website a long way from where I began and gave it a phenomenal look.

Also used Microsoft SQL Management Studio in order to support the database and create the tables needed in order to track the users that have registered to the website, it was also used to track the user's information (the amount of calories consumed that day, the day of the week, the workout that the user chose, and the completion of the workout selected) that has been logged down on the log page. There is also software built into visual studios that allows you to

Final Project Design and Explanation:

The web app launches with the home screen, this will be the screen that will give a brief overview of what the site has to offer to its users (see Figure 1).

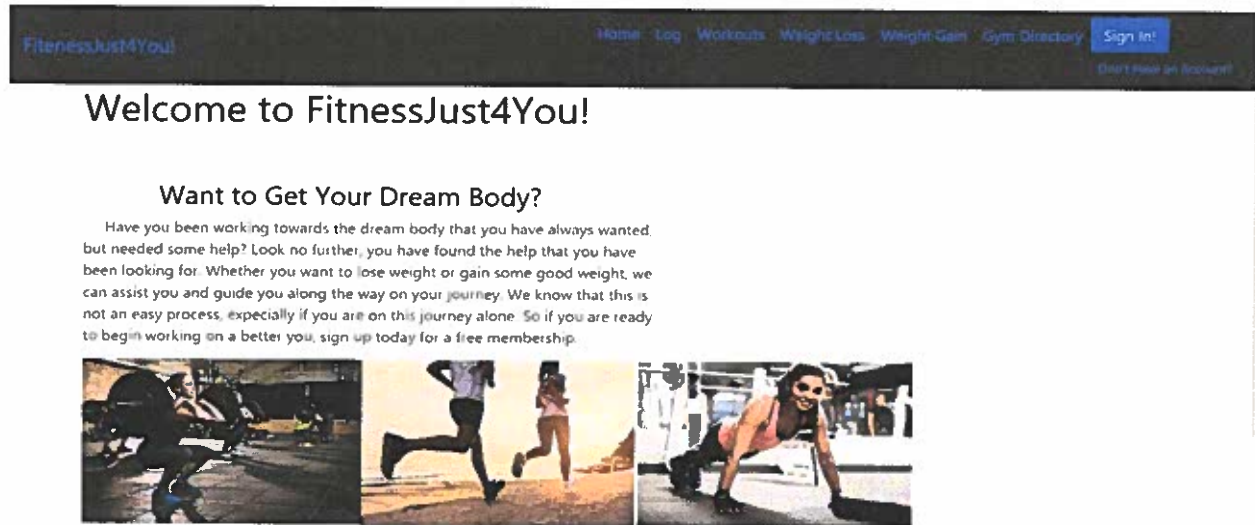


Figure 1. The home screen.

The next option on the task bar is the log page, this is where the user will navigate to in order to log down their caloric consumption, the day of the week, the exercise that they selected (if they did workout that day), and the percentage of the workout that was completed (See Figure 2).

There are quite a few software and hardware requirements for this project, but it was a great learning experience to expand my knowledge.

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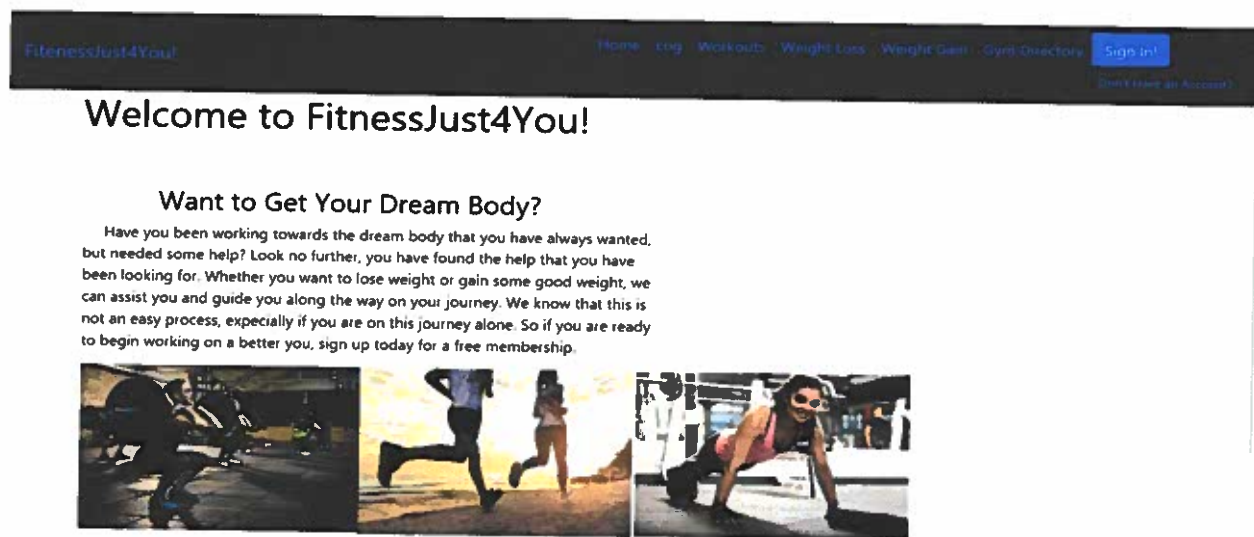


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FitnessJust4You! Home Log Workouts Weight Loss Weight Gain Gym Directory Sign In! Don't Have an Account?

LogJust4You!

Here is where you will be logging in and keeping track of your calorie intake, selecting which workout it is that you have chosen, to do that day, and fill in the amount of exercise for each day (by filling in a percentage of the workout that you completed). For the new comers, as you can see there is an area for your calorie intake, the day of the week, which workout you selected to do ("other" is if you chose to do your own workout), and the percentage of the workout completed. There will also be a box that will contain the intensity of your workout, whether it was a light workout, moderate workout, intense workout, or very intense workout. These categories are split up depending on how many calories you burned during the workout.

Total Calorie Intake Day of the Week Select Workout Workout Completion %

Submit

(Figure 2) The log page.

The error message below will pop up if a non-member or a member who has not logged in tries to log down information and store it. This page is only displayed before the user signs up and signs in so that they can see the full layout of the website, but this feature will not be accessible until the user is a member and is signed in (See Figure 3).

FitnessJust4You! Home Log Workouts Weight Loss Weight Gain Gym Directory Sign In! Don't Have an Account?

LogJust4You!

Here is where you will be logging in and keeping track of your calorie intake, selecting which workout it is that you have chosen, to do that day, and fill in the amount of exercise for each day (by filling in a percentage of the workout that you completed). For the new comers, as you can see there is an area for your calorie intake, the day of the week, which workout you selected to do ("other" is if you chose to do your own workout), and the percentage of the workout completed. There will also be a box that will contain the intensity of your workout, whether it was a light workout, moderate workout, intense workout, or very intense workout. These categories are split up depending on how many calories you burned during the workout.

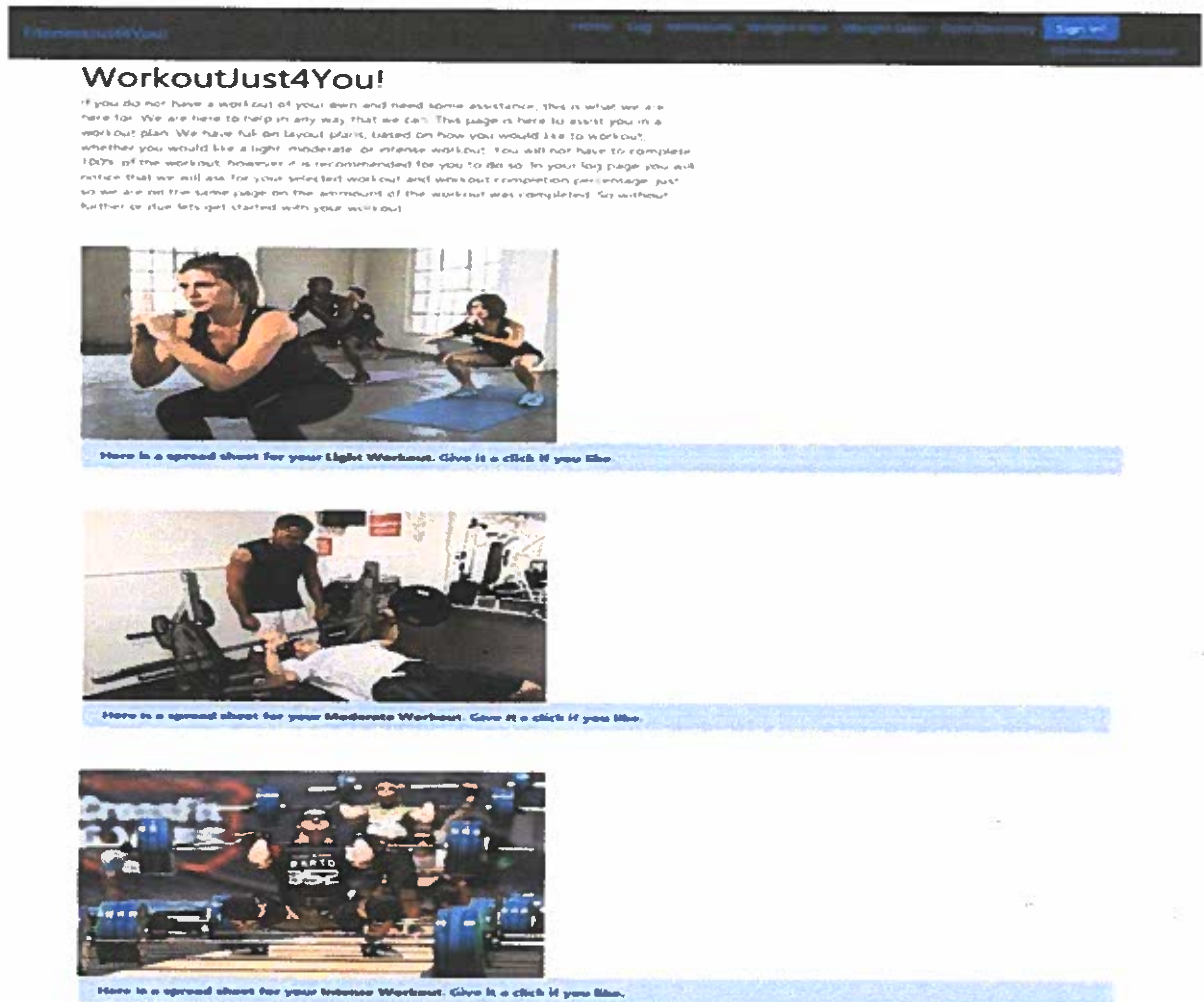
Total Calorie Intake Day of the Week Select Workout Workout Completion %

Submit

You Must Be a User To Log Information

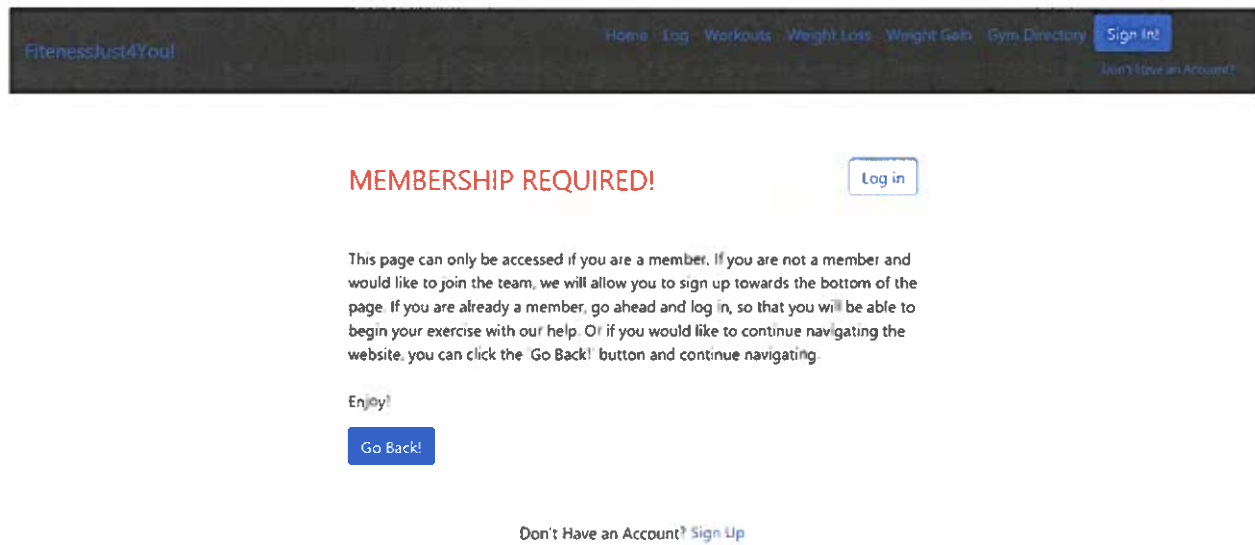
(Figure 3) Log Page Error Message

The next page will help the user if they are not experienced with creating their own workouts, we will assist them with workouts that have already been created. As you can see the user will have the option to choose from a light, moderate, or intense workout (See Figure 4).



(Figure 4) The Workout Selection Page

Below is the message that will appear on the screen if a user who is not a member or a user who has not logged in yet, tries to access the workout spreadsheets. It will display that the membership is required and have a short description (See Figure 5).



(Figure 5) Membership Required Message

Next, we have our weight loss page which is designed to help our users with their weight loss goals, whether it is through dieting and exercising or just sticking to a strict diet and losing weight without exercise. We have an outline with specific foods listed down for our users to follow (See Figure 6).

WeightLossJust4You!

Are you looking to lose that unwanted fat that you are sick of having at? Well we are here to help you on your journey to your goal. Whether you are looking to lose the weight with or without exercise we will help you with your goal. But you must stick to the meal that we are providing you with. So your diet will consist of the following:



Without Exercise:

If you plan to diet without exercise (whether it just your preference or if you are physically not able to exercise) your meal planning will be slightly more strict than if you did plan to exercise. So as your diet you MUST avoid all white carbohydrates such as rice, bread, cereal, pasta, potatoes, and of course any fried foods. With these rules your meals will become very repetitive, but you must stick with it in order to reach your goal. So your meal will consist of a lot of the following:

Proteins:

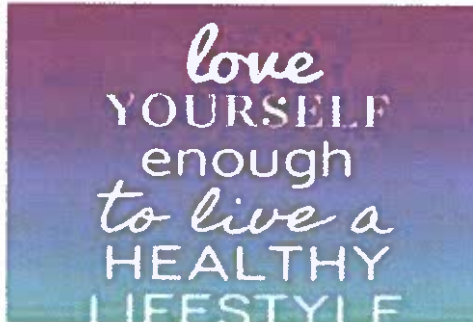
Egg whites (One whole egg to provide flavor)
 Chicken breast or thigh
 Grass fed Organic beef
 Pork

Legumes:

Lentils
 Black beans
 Pinto beans

Vegetables:

Squash
 Asparagus
 Peas
 Mixed Vegetables



Refrain from drinking calories, such as things like:

- Milk
- Normal soft drinks
- Fruit juice

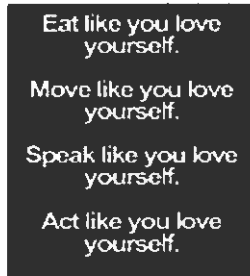
However you are able to drink the following:

- Water
- Unsweetened iced tea
- Coffee (without cream)
- Any other no/low calorie drink

Now, like stated before meals will be very repetitive, but you must stick to them. We will ask that you break your meals into four meals a day into the following layout:

- Breakfast: 7am - 10am
- Lunch: 1pm
- Snack/Lunch: 5pm
- Dinner: 9pm - 11pm

I may sound like a broken record, but it's for your good. You must make sure that you stick to this meal plan if you are choosing not to exercise during your weight loss journey. I know it can become very redundant, so you will actually be able to have one day a week, where you will not have to abide by the strict meal plan. You will be able to eat whatever it is that your stomach desires. The best to choose for this, would be on Saturdays. So I hope that this meal plan will help you reach your goal weight!



(Figure 6) The Weight Loss Page

After the weight loss webpage, we have our weight gain page designated to help our users with their goals in gaining weight, however we will not encourage packing on pounds the wrong way. A good diet and exercise are encouraged to ensure that while the user is gaining weight, they are putting on body mass versus body fat (See Figure 7).

WeightGainJust4You!

Help! There have you been trying to put on some weight, but you want to put on the right type of weight? Here I will help you on your weight gaining adventure and helping you gain your weight the right way. You will have to follow a some strict meal plan, but it will not be as strict as if you were trying to lose weight. You will need to follow the following:

BREAKFAST:

Glass of Orange Juice (12oz)

Bowl of cereal (preferably cheerios) with grape's bar

2 slices of wheat toast (with 1 tablespoon of peanut butter on each slice)

OR

Glass of Juice (12oz)

Breakfast Sandwich with English Muffin

2 Eggs

2 slices of ham

NO ONE IS PERFECT.
LOVE YOURSELF
AS YOU ARE
AND TRAIN HARD
TO BECOME
THE BEST THAT
YOU CAN BE.

© 2014 OPTIFAST

SNACK: (AFTER WORKOUT)

Sports drink: Pre workout or Mass gainer

With either 1 of the 4

Peanut butter and jelly sandwich

Container of yogurt with granola

Small bag of trail mix

Energy bar



LUNCH:

Lemonade or other juice (No soda)
Subsandwich (Foot long)

With choice of one meat:

- Chicken
- Steak
- Tuna
- Ham and cheese

Baked chips

or

Lemonade or other juice (No soda)
Grilled chicken salad
Baked potato

or

Lemonade or other juice (No soda)
Omelet (3 eggs + three vegetables)
Hashbrowns
2 slices of toast

SNACK:

Lemonade or other juice

With either 1 of the 3

- Banana with peanut butter
- Trail mix with cereal, nuts and dried fruits (2 handfulls)
- Cereal (large bowl)

DINNER:

Lemonade or other juice

Protein (CHOOSE ONLY ONE & DO NOT FRY)

- Steak
- Chicken
- Fish
- Pork (cut the fat off)
- Turkey

Carbohydrates (CHOOSE ONE)

- Pasta
- Baked potato
- Rice
- Corn

Side (CHOOSE ONE)

- Salad
- Cooked vegetables
- Fruits

SNACK:

Bowl of frozen yogurt or ice cream

Smoothie or protein shake

Peanut butter and jelly sandwich

(Figure 7) The Weight Gain Page

The next page will help our local users find a gym near them. This page displays a few gyms and their information including their website, street address, and phone number (See Figure 8).

DirectoryJust4You!



Website: [Click Here to Visit the Website](#)
 Address: 214 51 James Ave
 Goose Creek, SC 29645
 Phone: (843) 272-4798



Website: [Click Here to Visit the Website](#)
 Address: 829 Orangeburg Rd
 Summerville, SC 29483
 Phone: (843) 271-0057



Website: [Click Here to Visit the Website](#)
 Address: 2140 Northwoods Blvd
 North Charleston, SC 29406
 Phone: (843) 553-2921



Website: [Click Here to Visit the Website](#)
 Address: 368 E 1st St
 Summerville, SC 29484
 Phone: (843) 283-1195



Website: [Click Here to Visit the Website](#)
 Address: 7800 Rivers Ave
 North Charleston, SC 29406
 Phone: (843) 200-6159

(Figure 8) The Local Directory Page

When clicking on the “Don’t have an account?” link either at the top of the page in the task bar or in the sign in page, it will bring the user to this form and allow them to sign up and be able to access the rest of the features of the site. After successfully signing up the user will then be directed to the sign in sheet (See Figure 9).

The screenshot shows the 'Sign up' page for 'FitnessJust4You'. The header is dark with the site name and navigation links: Home, Log, Workout, Weight Loss, Weight Gain, Sign Directory, and Sign In. The main content area is white and contains a 'Sign up' form. The form has the following fields: First name (with placeholder 'Enter First Name'), Last name (with placeholder 'Enter Last Name'), Email address (with placeholder 'Enter Email'), Gender (with a dropdown menu showing 'Male'), Username (with placeholder 'Enter Username'), Password (with placeholder 'Enter Password'), and Re-Enter password (with placeholder 'Re-Enter Password'). A blue 'Register' button is at the bottom of the form. A 'Log in' button is in the top right corner of the form area. At the bottom of the page, there is a link 'Have an account? Log in'.

(Figure 9) The Sign-Up Page

On this page the user will enter the proper credentials in order to sign in to the website.

If the user is not a member yet, there is a link at the bottom of the page that will redirect the user back to the signup sheet (as stated in the description of Figure 9). If a non-user attempts to sign in, they will not be able to and an error message will be displayed because they do not have any

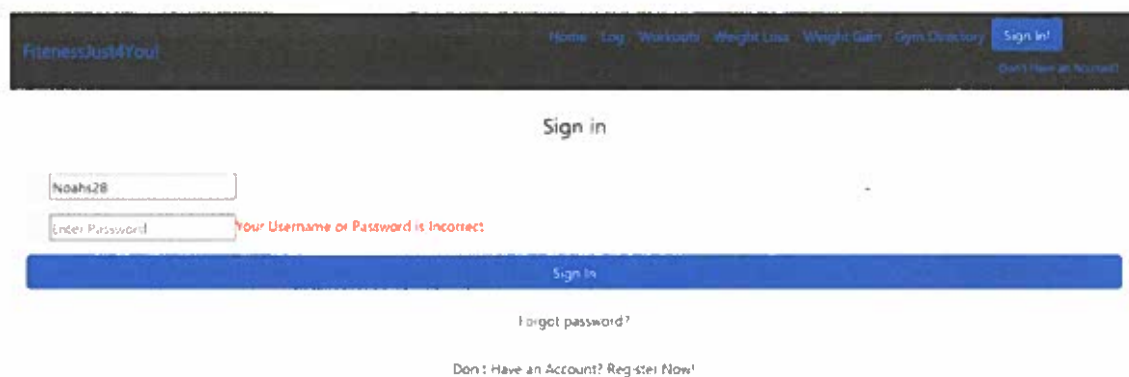
information stored within our database. If the log in is successful, they will be redirected back to the home page (See Figure 10).



The screenshot shows the 'Sign in' page of the FitnessJust4You website. At the top, there is a dark navigation bar with the site logo 'FitnessJust4You' on the left and a list of links: 'Home', 'Log', 'Workouts', 'Weight Loss', 'Weight Gain', 'Gym Directory', and a 'Sign in!' button. Below the navigation bar, the page title 'Sign in' is centered. There are two input fields: 'Enter Username' and 'Enter Password'. Below these fields is a large blue 'Sign in' button. Underneath the button, there is a link for 'Forgot password?'. At the bottom of the page, there is a link that says 'Don't Have An Account? Register Now!'.

(Figure 10) The Sign-In Page

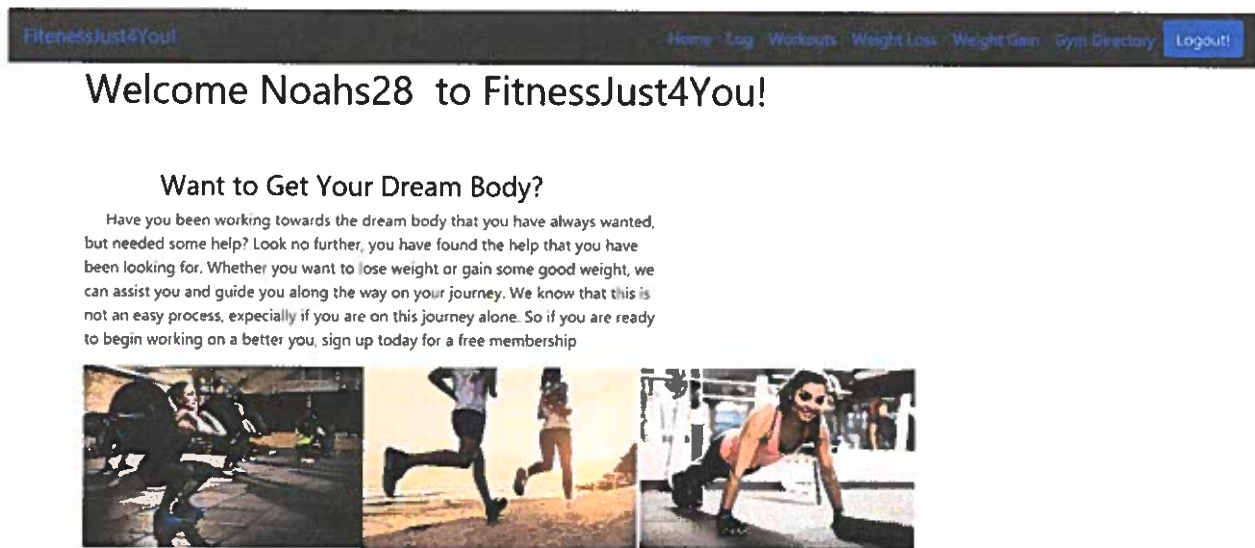
(See Figure 11) it is displaying the error message that will be displayed on the screen if any of these four conditions are met: not a member, entered wrong username, entered wrong password, entered wrong username and password



This screenshot shows the 'Sign in' page with an error message. The navigation bar is identical to Figure 10. The 'Sign in' title is centered. The 'Enter Username' field now contains the text 'Noahs28'. The 'Enter Password' field is empty. A red error message, 'Your Username or Password is incorrect', is displayed to the right of the password field. The large blue 'Sign in' button remains below the fields. The 'Forgot password?' link and the 'Don't Have An Account? Register Now!' link are still present at the bottom of the page.

(Figure 11) The Sign In Error Message

(See Figure 12) to view the layout of the home page once the user has logged in. As you can see this page will display the user's username at the top of the page. Since the user is logged in there is no longer a "Sign In!" button nor a "Don't Have an Account?" link. Since the user is now a member and logged in, there is now point in displaying those options instead, they have been replaced with a fully functional "Logout!" button.



(Figure 12) The Home Page Once Signed In

- The next figure you will see the user specific log page, as you can see it now says. "LogJust4Noahs28!" instead of "LogJust4You!". On this page the users will now be able to access the table and log down there their information because they are now a member. As you can see there is now a "See table" button at the bottom of the screen, that will redirect the user to the table that will display the information inputted into the log page.

FitnessJust4You
Home Log Workouts Weight Loss Weight Gain Gym Directory Logout!

LogJust4Noahs28 !

Here is where you will be logging in and keeping track of your calorie intake, selecting which workout it is that you have chosen, to do that day, and fill in the amount of exercise for each day (by filling in a percentage of the workout that you completed). For the new comers, as you can see there is an area for your calorie intake, the day of the week, which workout you selected to do (other is if you chose to do your own workout), and the percentage of the workout completed. There will also be a box that will contain the intensity of your workout, whether it was a light workout, moderate workout, intense workout, or very intense workout. These categories are split up depending on how many calories you burnt during the workout.

Total Calorie Intake
Day of the Week
Select Workout
Workout Completion %
Submit

See Table!

(Figure 13) The Log Page Once Signed In

Once the user has clicked on the “See Table” button they will be brought to this page which is displaying the log data table with the users input. This is where their information will be updated and stored, and they will be able to access it to see what they have done on previous days (See Figure 14).

FitnessJust4You
Home Log Workouts Weight Loss Weight Gain Gym Directory Logout!

LogTableJust4Noahs28 !

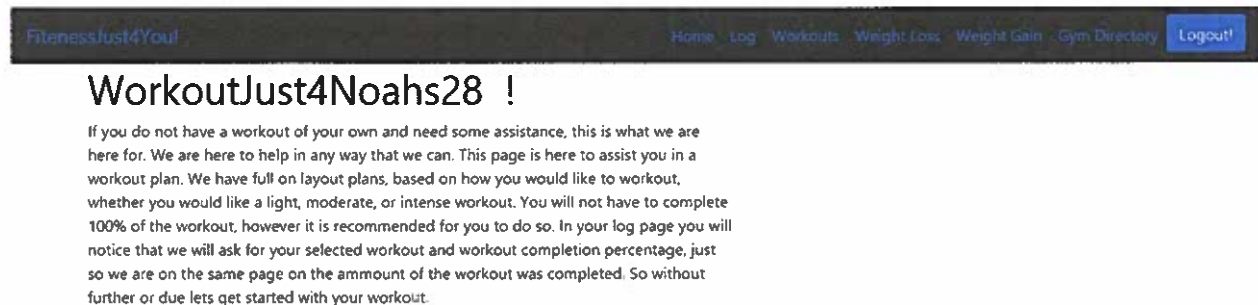
We want to help in any way possible that we can think of, so we have created this page for you ladies and gentlemen. Here is a table of the information you have previously logged down on your log page, it will be stored on this page for you to come back and visit whenever you please. Hope this helps you all! Enjoy!

Id	Calories	Day	Workout	Workout Completion
10	2000	Friday	Intense Workout	100
9	2000	Wednesday	Intense Workout	100
8	100	Wednesday	Intense Workout	75
7	1100	Friday	Moderate Workout	50
6	1100	Friday	Moderate Workout	50
5	100	Monday	Moderate Workout	75
4	100	Monday	Moderate Workout	50
3	100	Monday	Light Workout	75
2	75	Wednesday	Intense Workout	100
1	100	Monday	Light Workout	75

Go Back!

(Figure 14) The Log Information Table

Below is the user specified Workout page. The layout of the page has not changed, just the header of the page, now including the user's username. The other feature that has changed is that the user is now able to access the workouts because they have either signed up or logged in (See Figure 15). The same goes for Figure's 19, 20, and 21, as far as the layout not changing.



(Figure 15) The Workout Page Once Signed In

When the user clicks on the "Light Workout" they will see this layout of the light workout spreadsheet. Here is where they will go if they do not believe that they are physically fit or mentally ready for more intense workouts. Decided to go with a weekly workout spreadsheet, in order to help the users out. Didn't think that it was the best idea to give them one workout and possibly have them repeating the same workout every time they go to exercise, so this help them have more of a variety with their exercises. There is also a link at the bottom of the spreadsheet that will direct you back to the workout page. (See Figure 16).

[Home](#)
[Log](#)
[Weight Log](#)
[Weight Loss](#)
[About Us](#)
[Sign Up](#)

Light Workout Spreadsheet

Before you get started with the work out, make sure that you get a warm up in to ensure that your muscles are warm, loose and ready to work. Your warm up should be around 10-20 minutes long and should consist of stretching and dynamic warm up exercises that will get the blood flowing through your muscles more.

After you finish your workout be sure to stretch, so that you will not get the feeling of your muscles "locking up".

When looking through the workout, if you see a "/" that means that you will be alternating that workout, meaning when you get done with the first listed exercise go immediately into the next.

Monday

Number	Exercise	Sets	Reps
1	Push ups	5	10-15
2	Dumbell Curls	5	10-15
3	Body Weight Squats	5	10-15
4	Leg Extension	5	10-15
5	2 Min. Run (2mins running then 2mins walking on the Treadmill)	4	2Mins each
6	Abs (Core workout of your choice)	N/A	50

Wednesday

Number	Exercise	Sets	Reps
1	Pull ups (Assisted if needed)	5	10-15
2	Dips	5	10-15
3	Wall Sits	5	10-15
4	Leg Curls	5	10-15
5	Elliptical (Level 11 or above)	N/A	10Mins
6	Abs (Core workout of your choice)	N/A	50

Friday

Number	Exercise	Sets	Reps
1	Barbell Curls	5	10-15
2	Band Push Downs	5	10-15
3	Leg Press	5	10-15
4	Calf Raises	5	10-15
5	2 Min. Run (2mins running then 2mins walking on the Treadmill)	4	2Mins each
6	Abs (Core workout of your choice)	N/A	50

If you want to go back to the workout selection page, [Click Here](#)

(Figure 16) The Light Workout Spreadsheet

From the workouts page, if you click on the “Moderate Workout” link, it will bring you to the layout of the moderate workout spreadsheet. This page has an identical layout as Figure 16, but there will be additional workouts, which will bump up the intensity from light to moderate (See Figure 17).

Moderate Workout Spreadsheet

Before you get started with the work out, make sure that you get a warm up in to ensure that your muscles are warm, loose and ready to work. Your warm up should be around 10-20 minutes long and should consist of stretching and dynamic warm up exercises that will get the blood flowing through your muscles more.

After you finish your workout be sure to stretch, so that you **will not** get the feeling of your muscles "letting up".

When looking through the workout, if you see a " / " that means that you will be alternating that workout, meaning when you get done with the first listed exercise go immediately into the next.

Monday			
Repetition	Exercise	Time	Range
1	Bench Press	5	10
2	Wide Grip Push ups / Pushups or Chinups	5	10 / 10
3	Cable Pull Downs	5	10
4	Sequats	5	10
5	Glute Hams / Leg Extensions	5	10 / 10
6	Barly Weight Squat Airmen	5	10
7	20 Min Run (Treadmill or a track)	8	N/A
8	Abdo + Core workout of your choice!	N/A	1:30

Wednesday			
Repetition	Exercise	Time	Range
1	Military Press	5	10
2	Slight Bent Row / Bent Over Row	5	10 / 10
3	Lat Pull Downs	5	10
4	Front Squat	5	10
5	Dead Lift / Leg Curl	5	10 / 10
6	Calf Raises	5	10
7	Elliptical (level 11 or above)	N/A	20Mins
8	Abdo + Core workout of your choice!	N/A	1:30

Friday			
Repetition	Exercise	Time	Range
1	Incline Bench Press	5	10
2	Dips / Barnd Push (Downs)	5	10 / 10
3	Barbell Shrugs	5	10
4	Leg Press	5	10
5	Dumbbell Lunges / Kettlebell Swings	5	10 / 10
6	Good Mornings	5	10
7	2 Min Run (2mins running then 2mins walking on the Treadmill)	8	2Mins each
8	Abdo + Core workout of your choice!	N/A	30

(Figure 17) The Moderate Workout Spreadsheet

When clicking on the "Intense Workout" you will be directed to the layout of the intense workout spreadsheet, which is the last option that the users will be able to choose from. You will see that the exercises in this spreadsheet are far more intense than Figure 16 and Figure 17 (See Figure 18).

Intense Workout Spreadsheet

- Before you get started with the workout, make sure that you get a warm up in to ensure that your muscles are warm, loose and ready to work. Your warm up should be around 10-20 minutes long and should consist of stretching and dynamic warm up exercises that will get the blood flowing through your muscles more.
- After you finish your workout be sure to stretch, so that you will not get the feeling of your muscles "locking up".
- When looking through the workout, if you see a "+" that means that you will be alternating that workout, meaning when you get done with the first listed exercise go immediately into the next.

Monday			
Reps/Time	Exercise	Time	Reps
1	Bench Press / Dips	5	8-6-4-2-8 / 15
2	Water Grip Pull-ups / Marine Chest Fly	5	15 / 15
3	Dumbbell Shoulder/ Lateral Raise / Pulls	5	15 / 15
4	Squats / Leg Curls	5	8-6-4-2-8 / 15
5	Weighted Wall Sits / Leg Extensions	5	15 / 15
6	Weighted Split Leg Squat Lunges / Weighted Walking Lunges	5	15 / 15
7	2 Mile Run (Treadmill or a track)	1	N/A
8	Additional workouts of your choice	N/A	100

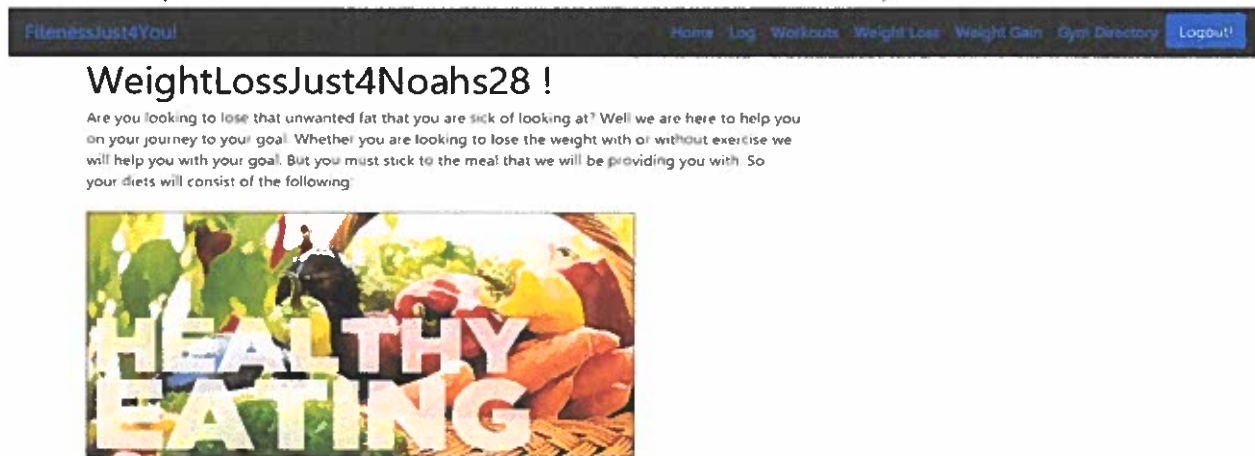
Wednesday			
Reps/Time	Exercise	Time	Reps
1	Power Clean / Weighted Bar Jumps	5	8-6-4-2-8 / 15
2	One Arm Dumb Press / Seated Lat Pull Downs	5	15 rep Arm / 15
3	Barbell Curls / Barbell Shrugs	5	15 / 15
4	Front Squat / Weighted Hops	5	8-6-4-2-8 / 15
5	Dead Lift / Lying Leg Curls	5	15 / 15
6	Dumbbell Squats / Calf Raises	5	15 / 15
7	Interval 15sec Sprinter	10	15 Seconds Each
8	Additional workouts of your choice	N/A	100

Friday			
Reps/Time	Exercise	Time	Reps
1	Inverted Bench Press / Band Pull Aparts	5	8-6-4-2-8 / 15
2	Dips / Band Push Downs	5	15 / 15
3	Barbell Overhead Squat / Plate Front Raise	5	10 / 15
4	Barbell Overhead Squat / Single Leg Curls Raises	5	8-6-4-2-8 / 15
5	Bulgarian Split Squats / Kettlebell Swings	5	15sec Rep / 15
6	Broad Jump / Med Ball Throw / Med Ball Throws (for Height)	5	15 / 15
7	2 Miles Run (Hill running then 2mins walking on the Treadmill)	8	2Miles each
8	Additional workouts of your choice	N/A	50

If you want to go back to the workout spreadsheet page [Click Here](#)

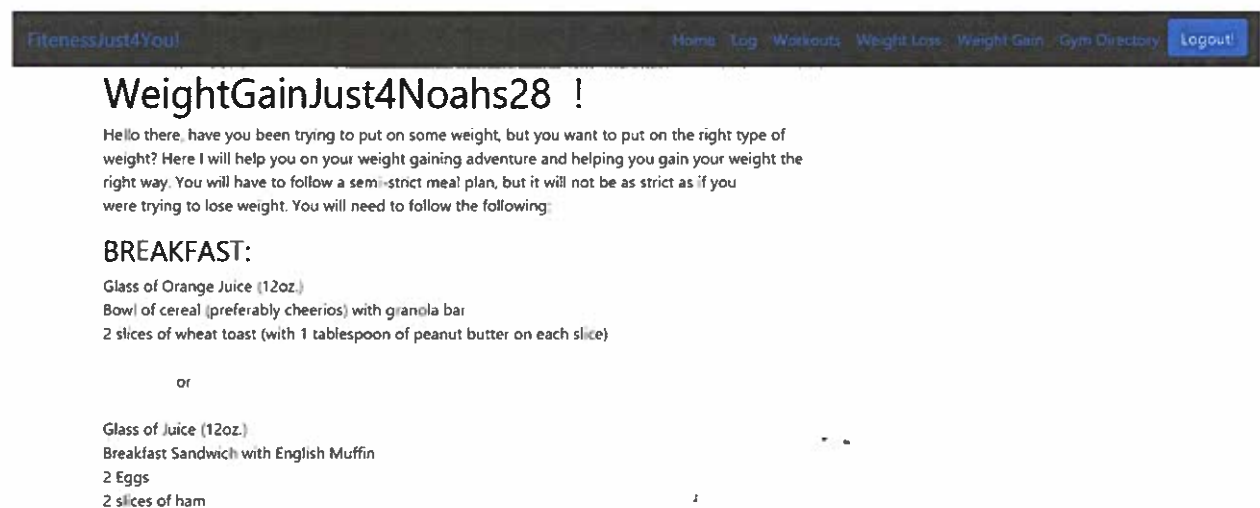
(Figure 18) The Intense Workout Spreadsheet

The next figure will show the user specified weight loss page. As stated before, the layout of the page does not change, just the header of the page (See Figure 19).



(Figure 19) The Weight Loss Page Once Signed In

Next you will see the display of the user specified weight gain page (See Figure 20).



(Figure 20) The Weight Gain Page Once Signed In

Lastly you will see the user specified local gym directory page (See Figure 21).

DirectoryJust4Noahs28 !



Website: [Click Here to Visit the Website](#)

Address: 214 St. James Ave,
Goose Creek, SC 29445
Phone: (843)737-4981

(Figure 21) The Local Directory Page Once Signed In

Test Plan:

Introduction:

- I will be testing my project through usability testing, as I will gather some individuals to run the website and provide feedback on how it functions and unit testing through visual studios.
- Goals and objectives
 - To have a properly functioning website, that can allow a user to register, sign-in (if registered already), log down their calorie intake and their exercise intensity for that day.
 - I believe the two options of testing that I have selected will give back the best feedback in order for my website to be the best
 - To my knowledge right now, I should not have any constraints.

References:

- Description of Project:
 - I will be creating a website for tracking calories for my project. I want this website to be available for the user to log down what they plan to eat ahead of time for breakfast, lunch, and dinner. First it will be able for the user to be able to register for an account, in order for them to be able to retrieve past logs that they have made. Once the user has his/her account he/she will be able to sign in using either their username and password or their email address and password. There will be some standard options that they may choose from, but also if what they ate did not appear in the list then they will have the option to it in manually. The website will allow users to set workout and calorie intake goals. The user will enter an example of what they plan to eat and their planned calorie intake. The user will then log how many calories he/she has eaten. The website will calculate how many calories the user is above or below his or her goal. Following that step, the user will log in the intensity of the day's workout. Users will enter

an intensity level of light, moderate, or intense workouts; these workout options will have a range of number (# of calories burned), which will help the user decide which category to select after a workout. If the customer has trouble finding workouts then there will also be a category there to help with that, it will have a few recommended workouts for the user to choose from.

- Propose Implementation Language(s):
 - I will be using Visual Studios (HTML) and javascript to implement the code. I will then be using CSS to make it look presentable on cell phone devices. SQL Database in order to store the user's information whenever they log in their information. And finally, I will use C# in order to write the server-side code in order to send the information to the user.
- Motivation and Problem Statement:
 - This project is important because there are a lot of people who want to track their calorie intake, but either don't know how to or don't have anywhere to log it down. Obesity has been an ongoing issue that has been addressed and there are individuals taking the steps that they need to handle the situation, but still might need some help, this will help this will help individuals learn how many calories that they are taking in with what they are eating on the regular. And what kind of workout they would need to do in order to either sustain their weight or either lose weight.
- Outline of Future Research Efforts:
 - I will be planning on finishing the remainder of the project on a weekly basis, as listed in the schedule. I feel that I am in good position to complete all of the tasks that I will need, in order to finish up. I believe that I have great knowledge in order to complete these tasks and my knowledge will still be improving through courses and research. I will continue to look in to more information in how I will be able to make the project as best as I can possibly make it.

Test Items:

- The web server
- The user input information
- The database
- Deployment to a public server
- Front-End UI
- Back-End UI

Features to be Tested:

- Navigate through the different web pages in the website
- Register for an account properly
- Able to sign-in once account is made
- Properly log calorie intake and exercise level.
- Make sure the database works properly, storing the information entered by the user.
- Make sure the users can click on the intensity level workout they want and receive recommended workouts.

- Make sure the links on the directory page properly send you to the correct gymnasium's website.

Features Not to Be Tested:

- A functioning mobile application.

Approach:

- The overall approach to the testing will be to gather a few users and see if they will be able to operate the website properly without any bugs. I will have the users testing the registration process, and once registered, will have them attempt to sign in. Then they will have the log page where they can log down their calorie intake and exercise intensity levels. They will also be allowed to receive some recommended workouts. They will also be allowed to retrieve the previous information that they have logged. And there will be a local gym directory with contact information and a link to their website.

Item Pass/Fail Criteria:

- If the user can click on each tab at the top of the screen and navigate throughout the website properly, then that will determine a pass for that portion of the project.
- If the user attempts to log down information on the log page without being a member or without being signed in and an error message is displayed under the submit button, this will determine a pass for that portion of the project.
- If the user can click on each "Sign Up" and "Sign In" link/button and it directs them to the correct pages, this will determine a pass for that portion of the project.
- If the user attempts to access the workout spreadsheets without being a member or without being signed in and it redirects them to a message saying that a membership is required, then that will determine a pass for that portion of the project.
- If the user can register, have their information stored and be directed to the sign in page, that will determine a pass for that portion of the project.
- If the user can sign in with the information that they have entered into the sign-up sheet, that will determine a pass for that portion of the project.
- When the user signs in, if they are redirected to the home page and their username is displayed at the top of the screen in the header (of each page), this will determine a pass for that portion of the project.
- If the information that has been stored on the log page, can be stored, recovered, and displayed into a table so the user can later recall their logged information, this will determine a pass for that portion of the project.
- Once logged in, if the user is able to click on the light, moderate, or intense button and receive a workout spreadsheet that they can then complete, this will determine a pass for that portion of the project.
- If the user can click on the links on the local gym directory and it takes them to the proper website, then this will determine a pass for that portion.
- If the user can click on the logout button and it successfully logs them out, this will determine a pass for that portion of the project.

Suspension Criteria and Resumption Requirements:

- There will be no suspension criteria that will come with this project.

Test Environment:

- Visual Studios community 2017
- HTML
- SQL Server Database

Estimate:

- This project is not estimated to cost anything because I am creating this website through visual studio community which is a free software

Staffing and Training Needs:

- I will be using two friends that will run the test on the website to make sure it works properly.
- The only requirements are to know the basics of operating a computer.
- If needed (training), I will show you how to navigate a website.

Responsibilities:

- The responsibilities for me will be to conduct a properly functioning website and for my two guests, it would be to test my website and make sure it is functioning with no errors and if there are any problems, for them to please let me know what problems they ran into.

Risks:

- From my current knowledge there will be no notable risks.

Assumptions and Dependencies:

- The assumptions that have been made are that I believe I will have a properly functioning website.
- The dependencies will be the web browser that will be needed in order to test the website.

Test Results:

- If the user can click on each tab at the top of the screen and navigate throughout the website properly, then that will determine a pass for that portion of the project.
 - PASS
- If the user attempts to log down information on the log page without being a member or without being signed in and an error message is displayed under the submit button, this will determine a pass for that portion of the project.
 - PASS
- If the user can click on each “Sign Up” and “Sign In” link/button and it directs them to the correct pages, this will determine a pass for that portion of the project.
 - PASS

- If the user attempts to access the workout spreadsheets without being a member or without being signed in and it redirects them to a message saying that a membership is required, then that will determine a pass for that portion of the project.
 - PASS
- If the user can register, have their information stored and be directed to the sign in page, that will determine a pass for that portion of the project.
 - PASS
- If the user can sign in with the information that they have entered into the sign-up sheet, that will determine a pass for that portion of the project.
 - PASS.
- When the user signs in, if they are redirected to the home page and their username is displayed at the top of the screen in the header (of each page), this will determine a pass for that portion of the project.
 - PASS
- If the information that has been stored on the log page, can be stored, recovered, and displayed into a table so the user can later recall their logged information, this will determine a pass for that portion of the project.
 - PASS
- Once logged in, if the user is able to click on the light, moderate, or intense button and receive a workout spreadsheet that they can then complete, this will determine a pass for that portion of the project.
 - PASS
- If the user can click on the links on the local gym directory and it takes them to the proper website, then this will determine a pass for that portion.
 - PASS
- If the user can click on the logout button and it successfully logs them out, this will determine a pass for that portion of the project.
 - PASS

Challenges Overcome:

Throughout the duration of the project, I have overcome a lot of challenges during the span of this project. The first problem that I encountered, was towards the end of my CSCI 498 course. With less than a month to go, my laptop crashed on me. This was very troubling time because it was the end of the semester and I used my laptop for all of my classes, not only for my project. I had to try and recover all of the progress that I have made throughout the entire

semester within three and a half weeks. Through a lot of long nights and a lot of focus, I was able to obtain the goals and make good progress on my project for a passing grade.

From there I was hoping for an easy process to make progress on the project, unfortunately that is not what happened. The next challenge that I had to deal with took place at the beginning of this semester (January 2019). On the laptop that I restarted my project on, I could not gain access to the c# tools that I needed to further implement my project. I went through Microsoft Visual Studios Installer, installed all of the c# options that I was given, but was still unable to access the tools. I then had to transfer over my project from the current laptop to my brand-new laptop that I have just purchased for another. When I transferred the project over, it did not include my latest updates that I made to the project. I saved the project plenty of times on a flash drive before I transferred over the project. When I would plug in my USB drive to the new laptop, the last saved date would be correct, but the recent updates that I have made to my project would not transfer over.

I had about two months' worth of progress made to my project that I lost during that transfer. After that obstacle, I just knew that everything after that should run smoothly, yet again I was let down. The third challenge that I had to overcome was not as drastic as the previous challenges, but none of the less, it's still an obstacle. I had no previous knowledge of bootstrap, before being told by my advisor to implement it into my project. Through a lot of research, I was able to learn the information and incorporate it into my project.

The project is now coming along very well and it's beginning to look more professional. The next challenge that I had to overcome, occurred the week before spring break of this year (February 21, 2019). Now working on the third laptop for this project, I have been only working on this machine for a month. My laptop then crashes, the second laptop that has crashed on me

during this project. I have now lost all of my progress again, so from February 22, 2019 to now I have been working non stop to have a presentable and functional project to demonstrate for my class.

Now, I have been working profusely to make great progress on my project to get ready for my presentation. Unfortunately, the challenges have not stopped, the last challenge that I have ran into were some minor challenges. While attempting to get the database to work I ran into some minor issues. Figuring out which database was more efficient to use, the local database or to connect to a database through Microsoft SQL Server Management Studio. I began to have some difficulty getting the databases to store the user's input into the database. The database would be properly connected but would not store any of the information. But by the grace of God, I was able to overcome all of the obstacles and have a fully operational project.

Future Enhancements:

- I have quite a few future enhancements that I would like to apply to my project. Here are the ideas that I have in mind:
 - Add a database that will store recipes and along with those recipes, store the number of calories you will consume if you are to eat that specified meal
 - Also store full meal layouts that the user can chose from.
 - Enhance the layout and the visuals of the website, so that it will look more acceptable for professional use.
 - Add a built-in calculator so that the user will simply be able to add in the meal that they ate that day and the number of calories consumed will be displayed to the user versus the user having to calculate their calories manually.

- Also be able to calculate the number of calories burnt when the percentage of a specific workout is entered in and be able to display that number back to the user, so that the user will know how much they have burned after the consumption of their meal.
- Add more to the directory page so the website will not only be beneficial to users in this specific local area.
 - Have a prompt that will ask the user for their location and display gyms in a specified range of that individual, so that gym's over a certain mile away will not displayed on their directory webpage.
- Incorporate weekly updates to the workouts, so the users are not repeatedly going through the same workout every week.
- Allow the website to operate as a mobile application.

Final Defense Presentation Slides:



(Slide 1)

1) Statement Of Purpose

- > As a former student athlete here at CSU I have decided to create a web application that can help its user with:
 - > Diet planning
 - > Tracking caloric intake
 - > Workouts
 - > Help users find a gym near them
- > When I began the project, the obesity rate was near 40% and raising.
 - > There are different levels of workouts stored within the website
 - > We know everyone is not on the same level physically and mentally
 - > Decided to go with this method so no one is discouraged.

(Slide 2)

2) Research/Background

- > I have had to conduct a lot of research throughout this project
 - > Had previous knowledge in HTML, CSS, and Databases
 - > Prior to this project, I had no knowledge of bootstrap
 - > YouTube
 - > www.w3schools.com
 - > www.getbootstrap.com

(Slide 3)

3) Hardware/Software Required for Project

- Required Hardware
 - The server (my laptop)
 - Power supply units
 - External hard drive (purchased after the third computer failure).
- Required Software
 - Visual Studios 2017
 - C#
 - Bootstrap4
 - HTML
 - CSS
 - JAVASCRIPT
 - Microsoft SQL Management Studio

(Slide 4)

4) Final Project Design

- I will now give you all a run through of how the project functions.

(Slide 5)

5) Test Plan / Results

- > Navigate through webpages properly
- > Non-members unable to access log table or workout spreadsheet
- > Databases work properly and store information
- > Sign up properly and be redirected to Sign In page
- > Properly log In
 - > Redirected to home page and have username displayed on webpages
- > Once logged in:
 - > Access log table
 - > Access workout spreadsheets
- > Gym directory links navigate you to correct website
- > Logout button navigates user back to the beginning stage.
- > All test cases have passed!

(Slide 6)

6) Challenges Overcome

- > Laptop #1 Crash
- > Laptop #2 / C# tools
 - > Transfer project to laptop #3
- > No previous knowledge of Bootstrap4
- > Laptop #3 Crash
 - > I would encourage future students to buy an exterior hard drive.
- > Database troubles

(Slide 7)

7) Future Enhancements

- > Expand database in order to store recipes, meal layouts, and the number of calories you will consume if you are to eat that specified meal
- > Enhance the layout and the visuals of the website, for a more professional look.
- > Add a built in calculator
 - > User enters in meals and number of calories automatically add up and displays back to user
 - > Also be able to calculate the number of calories burnt when the percentage of a specific workout is entered in

(Slide 8)

8) Future Enhancements Cont'd

- > Add more to the directory page so the website will not only be beneficial to users in this specific local area.
 - > Have a prompt that will ask the user for their location and display gyms within a specified range.
- > Incorporate weekly updates to the workouts.
- > Allow the website to operate as a mobile application.

(Slide 9)